GenHack 2025 - Week 1 - Warm-Up

Team 31 : Sbiyyaat

Introduction & Objectives

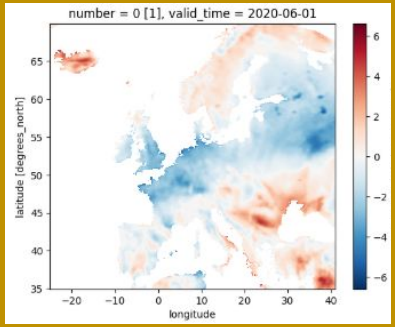
Objective : Load, inspect, and understand the three core datasets provided for the analysis of Urban Heat Islands (UHI) and Climate Downscaling.

1. **ERA5-Land (Modelled Climate Data)**
2. **Sentinel-2 NDVI (Vegetation Data)**
3. **ECA&D Ground Stations (Observed “Ground Truth”)**

Primary Data Sources

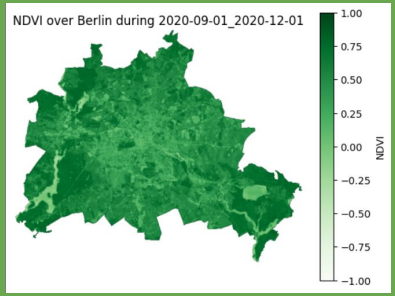
ERA5-Land

Gridded climate data that provides a consistent view of the evolution of land variables over several decades at an enhanced resolution compared to ERA5.



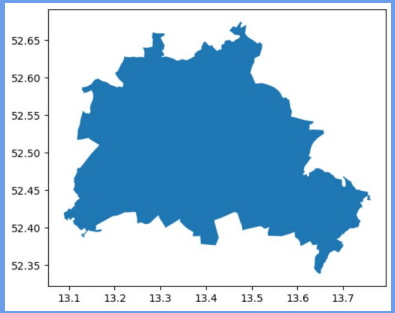
NDVI

Satellite imagery. It gives an index for quantifying green vegetation. Values range from -1 to 1. Loaded satellite imagery and clipped it to our Berlin geometry.



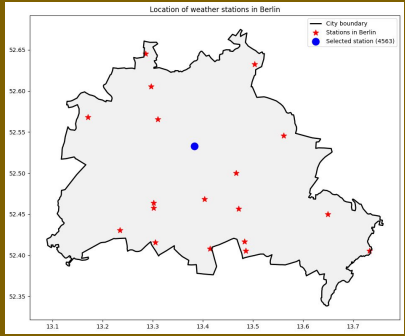
GADM

provides maps and spatial data for all countries and their sub-divisions.
For our case, we limited ourselves to European countries.



ECA&D

For Ground stations.
Loaded raw text files, filtered for valid datas.



Exploration 1: Gridded Data (ERA5 & NDVI)

This is the "big picture" data, showing models and satellite observations.

ERA5-Land (Temperature)

- **Type:** Gridded reanalysis data.
- **Resolution:** Coarse (e.g., 0.1° grid). One pixel covers a large, mixed area.
- **Finding:** Provides a complete, consistent map but "smooths over" local variations.

Sentinel-2 (NDVI)

- **Type:** Gridded satellite observation.
- **Resolution:** High (e.g., 10m).
- **Finding:** Clearly distinguishes vegetated areas (parks) from non-vegetated (urban centers). This will be key to explaining UHI.

Exploration 2 : Ground Truth (ECA&D Stations)

This is the "ground truth" data, showing precise, local measurements.

Case Study: Berlin

We filtered the GADM dataset to isolate Berlin.
We then spatially joined to find all 10+ ground stations located within the city limits.

Single Station Time Series (e.g., STAID 10382)

Action: Plotted daily max temperature (TX) for a single station.
Finding: This data is precise, observed, and captures the actual temperature at a specific point, which is essential for validation.

Key Results & Next Steps

- **The Core Problem: Model vs. Reality :**

Key Finding: Our initial exploration confirms the central challenge. The coarse ERA5 model data (blue line) does not match the precise Ground Station data (red line) for the same location.

This discrepancy is the problem we are here to solve. It's the "bias" caused by local effects (like UHI) that "Downscaling" aims to correct.

- **Next Steps (Moving to Period 2 & 3):**

1. **Visualize** (Week 2): Create detailed maps showing where this discrepancy is largest (e.g., city center vs. parks).
2. **Quantify** (Week 3): Develop metrics to measure this bias.
3. **Explain** (Week 4): Use NDVI data as an explanatory variable to model why this bias exists (Hypothesis: Low NDVI -> High Bias).